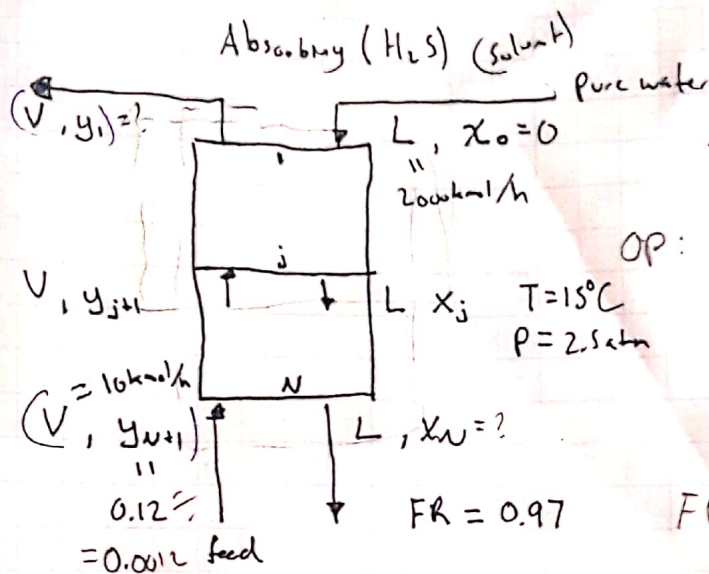


12.02

8.5



$$V y_{j+1} + L x_0 = V y_1 + L x_j$$

$$OP: y_{j+1} = \frac{L}{V} x_j + \left(y_1 - \frac{L}{V} x_0 \right)$$

$$FR = \frac{D x_{10}}{F z_L} \Rightarrow \frac{L x_N}{V y_{N+1}}$$

a) Calculate outlet gas & liquid mol fractions of H_2S
(y_1) = ? (x_N) = ?

$$DMB: V y_1 + L x_N = L x_0 + V y_{N+1}$$

$$V y_1 = V y_{N+1} - L x_N$$

$$L x_N = FR V y_{N+1}$$

$$y_1 = y_{N+1} - \frac{1}{V} (L x_N)$$

$$x_N = \frac{FR V y_{N+1}}{L}$$

$$= y_{N+1} - \frac{1}{V} (FR \cdot V y_{N+1})$$

$$x_N = \frac{(0.97)(10 \text{ kmol/h})(0.0012)}{2000 \text{ kmol/h}}$$

$$= y_{N+1} - FR \cdot y_{N+1}$$

$$y_1 = (1 - FR) y_{N+1}$$

$$= (1 - 0.97)(0.0012)$$

$$y_1 = 3.6 \times 10^{-5}$$

$$x_N = 5.82 \times 10^{-6}$$

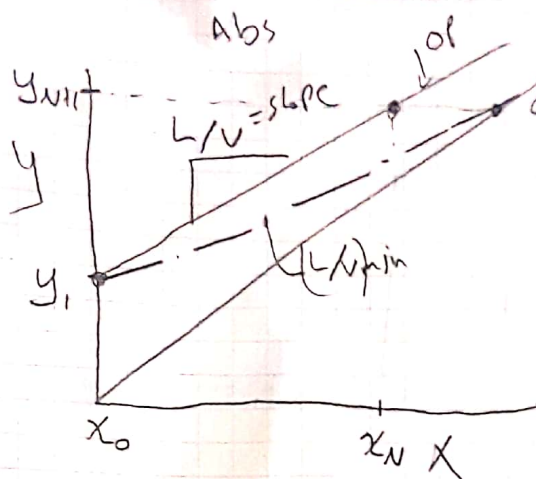
b) Calculate the # of ea. stages required using McCabe theta diagram

$$T = 15^\circ C \Rightarrow (H = 423)$$

$$y_B = \frac{(423 \text{ atm})(x_B)}{2.5 \text{ atm}}$$

$$y_B = \frac{P_B}{P_{tot}} = H \frac{x_B}{P_{tot}}$$

$$y_B = 169.2 x_B \quad \text{exact}$$



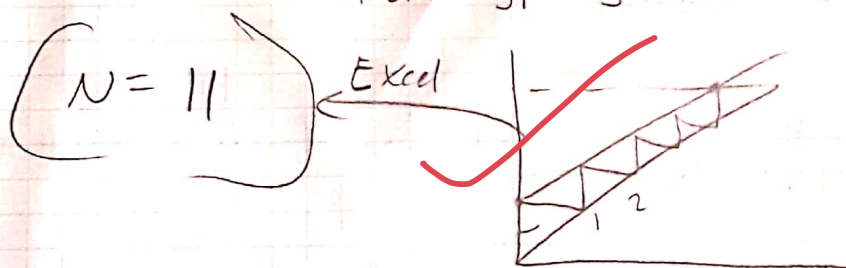
$$\frac{L}{V} = \frac{y_{N+1} - y_1}{x_N - x_0} = \frac{0.0012 - 3.6 \times 10^{-5}}{5.82 \times 10^{-4}}$$

$$\left(\frac{L}{V} = 200 \right) \quad (\text{or just } \frac{L}{V} = \frac{2000}{10})$$

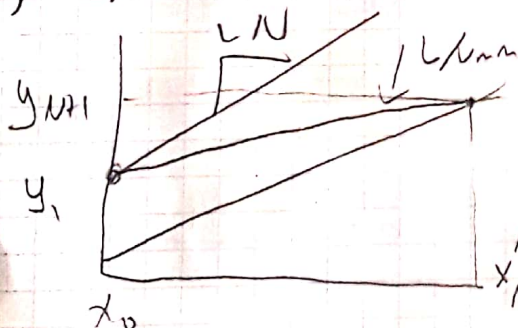
$$\text{op} \Rightarrow y_{j+1} = \frac{L}{V} x_j + \left(y_1 - \frac{L}{V} x_0 \right)$$

$$= \frac{L}{V} x_j + y_1$$

$N = ?$ From excel graph step off stages from $y_1 \rightarrow y_{N+1}$



c) $L/V = M (L/V)_{\min}$ $M = ?$



$$\left(\frac{L}{V_{\min}} \right) = \frac{y_{N+1} - y_1}{x_N^m} = \frac{0.0012 - 3.6 \times 10^{-5}}{\frac{0.0012}{164.2}}$$

$$(L/V)_{\min} = 164.124$$

$$x_N^A = y_0^A = y_{N+1}^A = \frac{y_0^A}{164.2} \Rightarrow \frac{L}{V} = \frac{200}{(L/V)_{\min}} = \frac{200}{164.124} = 1.2185$$

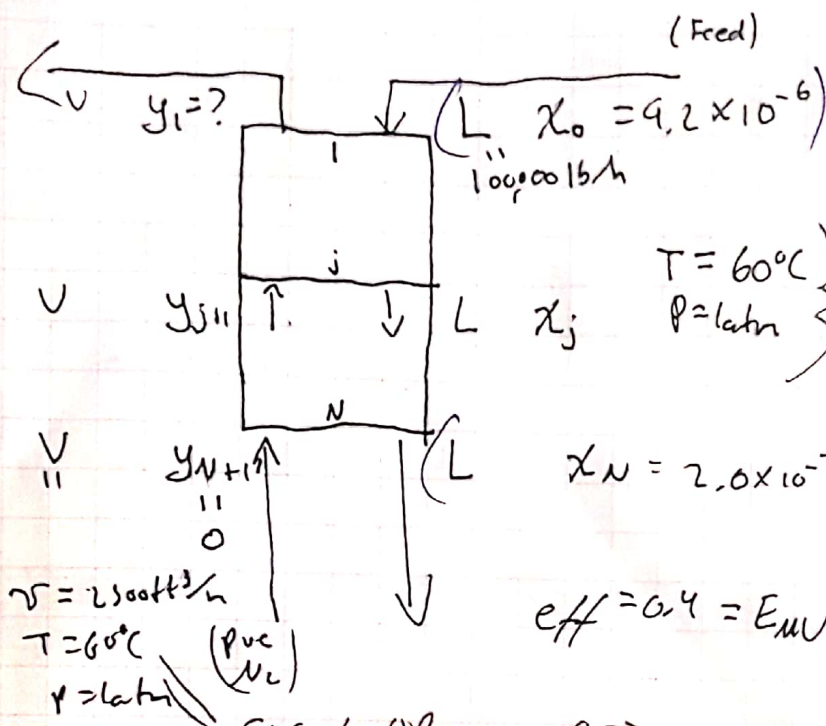
$$\left(M = 1.2185 \right)$$

Why is this not practical? what to do to make it practical?
The M value isn't high enough at 1.2185
 L/V pretty close to $(L/V)_{\min}$

Increase the L and flow rate

12.06

stripping (CO₂)



N = ?

$$\left. \begin{array}{l} T = 60^\circ\text{C} \\ P = 1 \text{ atm} \end{array} \right\} \Rightarrow y_B = \frac{P_B}{P_{\text{tot}}} = \frac{H_B x_0}{P_{\text{tot}}}$$

$$H_B = 3410$$

$$(y_B = 3410 x_0)$$

$$x_N = 2.0 \times 10^{-7}$$

$$\text{eff} = 0.4 = E_{\text{MU}}$$

(1) V = ?

Ideal Gas (M₂)

$$\rho_{M_2} = 28 \frac{\text{lbmol}}{\text{lbmol}}$$

$$PV = nRT$$

$$P_V = \frac{V}{V} RT$$

$$V = \frac{P_V}{RT} = \frac{(1 \text{ atm})(2500 \text{ ft}^3/\text{h})}{(0.73 \text{ atm-ft}^3/\text{lbmol-R})(600 \text{ R})} = 5.71 \frac{\text{lbmol}}{\text{h}}$$

$$(2) (L) = ? \quad 100,000 \text{ lb/h} \times \frac{1 \text{ lbmol}}{18 \text{ lb}} = 5555.55 \frac{\text{lbmol}}{\text{h}}$$

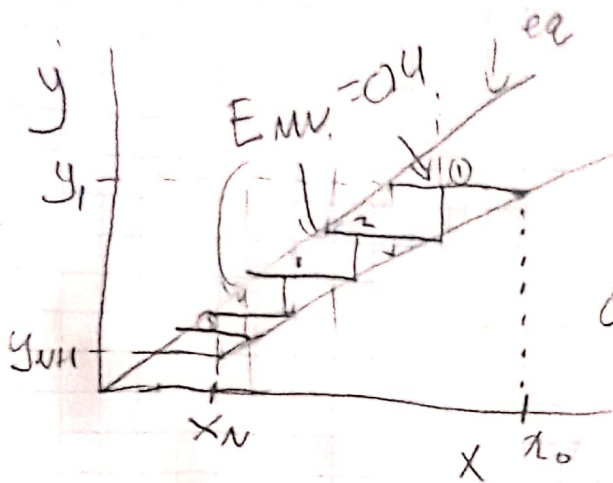
$$\text{OMB: } V y_{N+1} + L x_0 = V y_1 + L x_N$$

$$y_1 = \frac{V y_{N+1} + L x_0 - L x_N}{V}$$

$$= \frac{(5555)(4.2 \times 10^{-6}) - (5555)(2 \times 10^{-7})}{5.71}$$

$$y_1 = 0.008756$$

stripping



$\Rightarrow OP$

$$\Rightarrow y_{j+1} = \frac{L}{U} x_j + (y_1 - \frac{L}{U} x_0)$$

$$= \frac{5555}{5.71} x_j + (0.008756 - \frac{5555}{5.71} (0.008756))$$

$$OP \Rightarrow y_{j+1} = 972.8546 - 1.94 \times 10^{-4}$$

Note $\Rightarrow EMU = 0.4$
 $\Rightarrow (N = 5)$

From Excel graph w/ $EMU = 0.4$

$$m = \frac{y_1 - y_0}{x_1 - x_0} = \frac{y_1 - y_0}{x_1 - x_0} = 34 \text{ per}$$

b) kernel can $y_1^* = 3410 x_0 = 0.01311$

$$N = \frac{-\ln \left(\left(1 - \frac{m \cdot U}{L} \right) \left(\frac{y_{N+1} - y_1^*}{y_1 - y_1^*} \right) + \frac{m \cdot U}{L} \right)}{\ln \left(1 + EMU \left(\frac{m \cdot U}{L} - 1 \right) \right)}$$

$$= \frac{-\ln \left(\left(1 - 3410 \left(\frac{1}{972} \right) \right) \left(\frac{0 - 0.01314}{0.00876 - 0.01314} \right) + 3410 \left(\frac{1}{972} \right) \right)}{\ln \left(1 + 0.4 \left(\frac{3410}{972} - 1 \right) \right)}$$

$(N = 5)$ ~~X~~

$= 5.016$

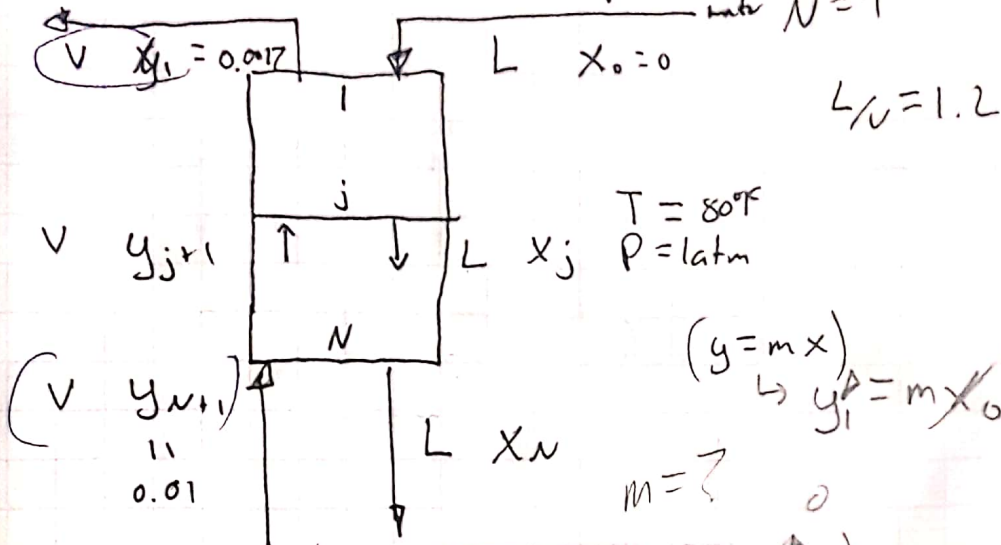
6 stages

- 0.5

D11

Absorption

Pure distilled water $N=4$



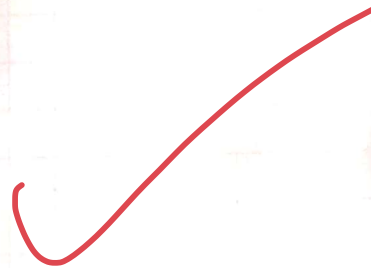
$$L/V = 1.2$$

$$\Rightarrow N = \frac{\ln \left[\left(1 - m \frac{V}{L} \right) \left(\frac{y_{N+1} - y_1^*}{y_1 - y_1^*} \right) + m \frac{V}{L} \right]}{\ln \left(\frac{1}{m} \frac{L}{V} \right)}$$

$$4 = \frac{\ln \left[\left(1 - m \frac{1}{1.2} \right) \left(\frac{0.01 - 0}{0.017 - 0} \right) + m \left(\frac{1}{1.2} \right) \right]}{\ln \left(\frac{1}{m} (1.2) \right)}$$

Goulbeck
50%

$$(m = 1.4128)$$



$$V \quad y_1 = 0.000008 \text{ meth} \quad L \quad x_0 = 0$$

$$V \quad y_{j+1}$$

$$L \quad x_j \quad T = 252.35 \quad P = 172 \text{ atm}$$

$$L/V = 26$$

$$x_N = ?$$

$$\begin{array}{l} V \\ \parallel \\ 100 \text{ kmol/h} \\ y_{NH_3} = 0.0002 \\ y_{Ar} = 0.0013 \\ \text{meth} \end{array}$$

$$P_M = 3600 \times x_N$$

$$P_A = 7700 \times x_{Ar}$$

$$y_A = \frac{7700}{175} x_{Ar} = 44 x_{Ar}$$

$$y_{meth} = \frac{P_M}{P_{tot}} = \frac{3600}{175} x_N = 20.5714 x_N$$

$$y_{meth} = 20.5714 x_N$$

$$y_1 = m(x_0) = 0$$

a) Mol. balance:

$$L x_0 + V y_{NH_3, meth} = V y_{1, meth} + L x_{N, meth}$$

$$x_{N, meth} = \frac{V y_{NH_3, meth} - V y_{1, meth}}{L}$$

$$= \frac{V}{L} (y_{NH_3, meth} - y_{1, meth})$$

$$= \frac{1}{26} (0.0013 - 0.000008)$$

$$(x_{N, meth} = 4.969 \times 10^{-5})$$

$$b) N = \ln \left(\left(1 - m \left(\frac{V}{L} \right) \right) \left(\frac{y_{NH_3} - y_1}{y_1 - y_1^*} \right) + \frac{mV}{L} \right) = 15.12$$

$$= \ln \left[\left(1 - 20.5714 \left(\frac{1}{26} \right) \right) \left(\frac{0.0013}{0.000008} \right) + 20.5714 \left(\frac{1}{26} \right) \right] = \ln \left(\frac{26}{20.5714} \right)$$

$$c) \frac{y_{n+1} - y_1}{y_1 - y_1} = \frac{L/mv - (L/mv)^{n+1}}{1 - L/mv}$$

$$\frac{y_{n+1}}{y_1} = 1 + \frac{L/mv - (L/mv)^{n+1}}{1 - L/mv}$$

NOTE (m = margin)

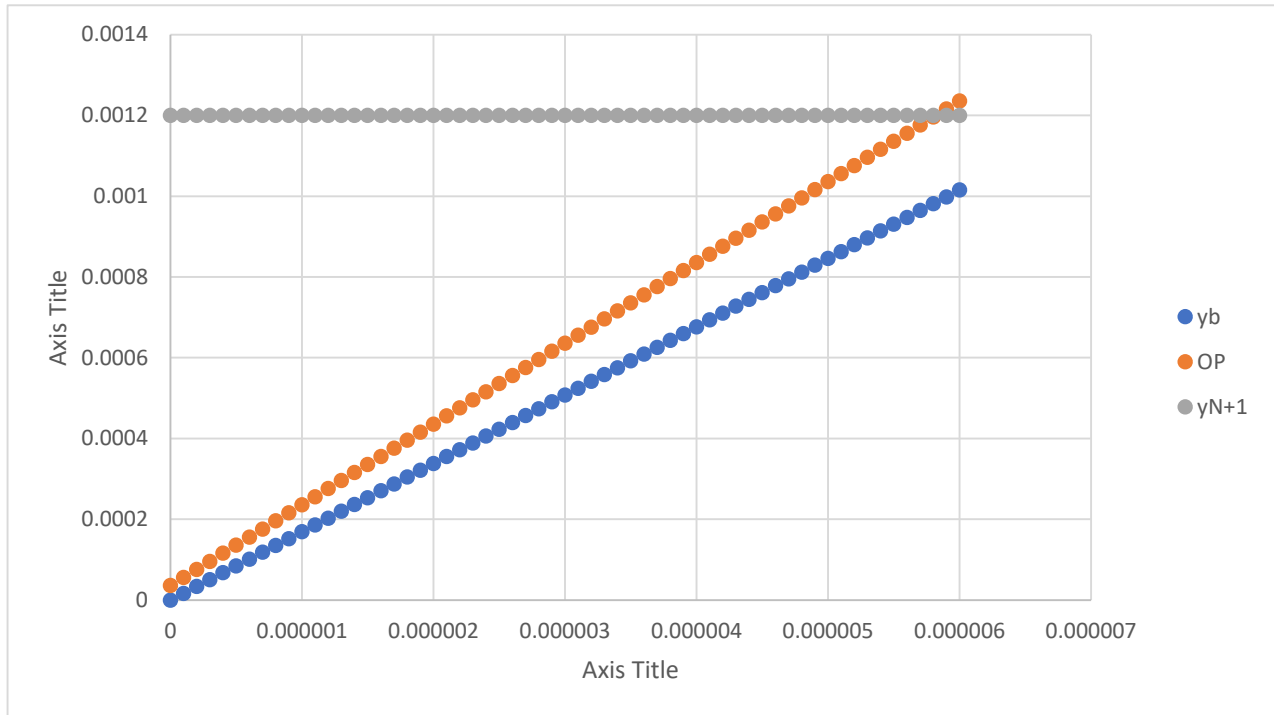
$$y_1 = \frac{y_{n+1}}{\left(1 + \frac{L/mv - (L/mv)^{n+1}}{1 - L/mv}\right)}$$

$$y_1 = \frac{0.0002}{1 + \frac{(26/444)^{15.156+1} - (26/444)}{1 - 26/444}}$$

$$y_1 = 8.183 \times 10^{-5}$$

xb	yb	OP	L/Vmin	yN+1
0	0	0.000036		0.0012
0.0000001	0.00001692	0.000056		0.0012
0.0000002	0.00003384	0.000076		0.0012
0.0000003	0.00005076	0.000096		0.0012
0.0000004	0.00006768	0.000116		0.0012
0.0000005	0.0000846	0.000136		0.0012
0.0000006	0.00010152	0.000156		0.0012
0.0000007	0.00011844	0.000176		0.0012
0.0000008	0.00013536	0.000196		0.0012
0.0000009	0.00015228	0.000216		0.0012
0.000001	0.0001692	0.000236		0.0012
0.0000011	0.00018612	0.000256		0.0012
0.0000012	0.00020304	0.000276		0.0012
0.0000013	0.00021996	0.000296		0.0012
0.0000014	0.00023688	0.000316		0.0012
0.0000015	0.0002538	0.000336		0.0012
0.0000016	0.00027072	0.000356		0.0012
0.0000017	0.00028764	0.000376		0.0012
0.0000018	0.00030456	0.000396		0.0012
0.0000019	0.00032148	0.000416		0.0012
0.000002	0.0003384	0.000436		0.0012
0.0000021	0.00035532	0.000456		0.0012
0.0000022	0.00037224	0.000476		0.0012
0.0000023	0.00038916	0.000496		0.0012
0.0000024	0.00040608	0.000516		0.0012
0.0000025	0.000423	0.000536		0.0012
0.0000026	0.00043992	0.000556		0.0012
0.0000027	0.00045684	0.000576		0.0012
0.0000028	0.00047376	0.000596		0.0012
0.0000029	0.00049068	0.000616		0.0012
0.000003	0.0005076	0.000636		0.0012
0.0000031	0.00052452	0.000656		0.0012
0.0000032	0.00054144	0.000676		0.0012
0.0000033	0.00055836	0.000696		0.0012
0.0000034	0.00057528	0.000716		0.0012
0.0000035	0.0005922	0.000736		0.0012
0.0000036	0.00060912	0.000756		0.0012
0.0000037	0.00062604	0.000776		0.0012
0.0000038	0.00064296	0.000796		0.0012
0.0000039	0.00065988	0.000816		0.0012
0.000004	0.0006768	0.000836		0.0012
0.0000041	0.00069372	0.000856		0.0012
0.0000042	0.00071064	0.000876		0.0012
0.0000043	0.00072756	0.000896		0.0012
0.0000044	0.00074448	0.000916		0.0012
0.0000045	0.0007614	0.000936		0.0012

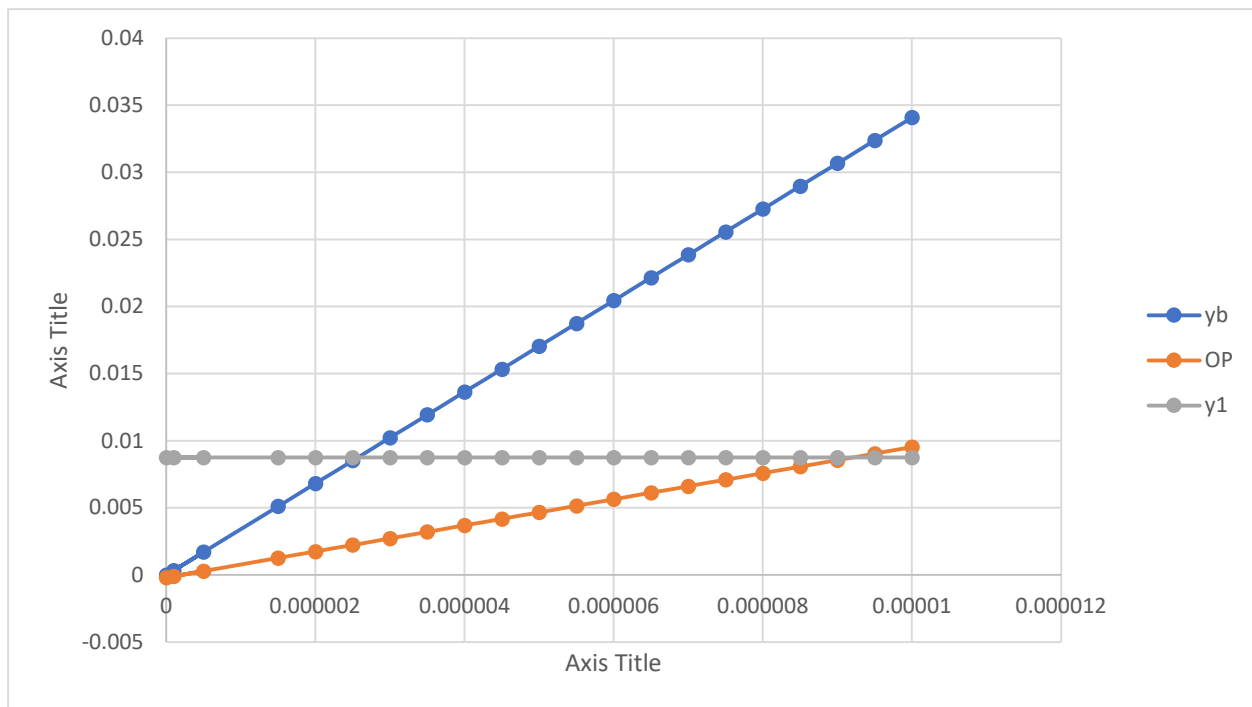
0.0000046	0.00077832	0.000956	0.0012
0.0000047	0.00079524	0.000976	0.0012
0.0000048	0.00081216	0.000996	0.0012
0.0000049	0.00082908	0.001016	0.0012
0.000005	0.000846	0.001036	0.0012
0.0000051	0.00086292	0.001056	0.0012
0.0000052	0.00087984	0.001076	0.0012
0.0000053	0.00089676	0.001096	0.0012
0.0000054	0.00091368	0.001116	0.0012
0.0000055	0.0009306	0.001136	0.0012
0.0000056	0.00094752	0.001156	0.0012
0.0000057	0.00096444	0.001176	0.0012
0.0000058	0.00098136	0.001196	0.0012
0.0000059	0.00099828	0.001216	0.0012
0.000006	0.0010152	0.001236	0.0012



draw stages on
graph !

-0.5

xb	yb	OP	L/Vmin	y1
0	0	0	-0.000195157	0.008756
0.0000005	0.001705	0.000291319		0.008756
0.0000001	0.000341	-9.78617E-05		0.008756
0.0000015	0.005115	0.001264271		0.008756
0.000002	0.00682	0.001750747		0.008756
0.0000025	0.008525	0.002237223		0.008756
0.000003	0.01023	0.002723699		0.008756
0.0000035	0.011935	0.003210175		0.008756
0.000004	0.01364	0.00369665		0.008756
0.0000045	0.015345	0.004183126		0.008756
0.000005	0.01705	0.004669602		0.008756
0.0000055	0.018755	0.005156078		0.008756
0.000006	0.02046	0.005642554		0.008756
0.0000065	0.022165	0.00612903		0.008756
0.000007	0.02387	0.006615506		0.008756
0.0000075	0.025575	0.007101982		0.008756
0.000008	0.02728	0.007588458		0.008756
0.0000085	0.028985	0.008074934		0.008756
0.000009	0.03069	0.00856141		0.008756
0.0000095	0.032395	0.009047886		0.008756
0.00001	0.0341	0.009534361		0.008756



Same, stages!

-0.5

4

4

1.412884